Large values of Dirichlet polynomials: Difference sets and additive energy

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1 Recap

Let $S \subseteq [0,T]$ be a 1-spaced set, and let $n \sim N$ denote the range $N < n \leq 2N$. In previous weeks, we have seen that estimates for the mean value

$$\sum_{t\in S}\Big|\sum_{n\sim N} a_n n^{it}\Big|^2$$

give rise to large value estimates for Dirichlet polynomials. In particular, we have seen that known bounds for the above give the following theorem:

Theorem 1 (Montgomery–Halász–Huxley large value estimate)**.** *Let $S \subseteq [0,T]$ be a 1-spaced set. Suppose that $(b_n)_{n\leq N}$ are complex numbers with $|b_n| \leq 1$, and $S_V \subseteq S$ are points $t \in S$ such that*

$$\Big|\sum_{n\sim N} b_n n^{it}\Big| \geq V,$$

Then we have

$$|S_V| \leq T^{o(1)}\Big(N^2V^{-2} + T\min\big\{NV^{-2}, N^4V^{-6}\big\}\Big). \tag{1}$$

This follows by combining the classical Mean Value Theorem with the estimate of Huxley [4] (which utilises a mean value theorem of Montgomery [6], building on Halász [2]). Note that when $V = N^{3/4}$, the estimates coming from the classical Mean Value Theorem and Huxley [4] agree. It is in this range that Guth and Maynard [1] get improvements. They begin by splitting into two cases:

1. The case where S_V has high additive energy[1].
2. The case where S_V does not have high additive energy.

The main innovations in their paper are in the handling of the second case. Here, we discuss how they handle the first case.

2 Mean values over difference sets

The key result used by Guth and Maynard [1] to handle the high additive energy case is the following:

[1]Here, we define the additive energy of a set W to be $E(W) = \#\{|t_1 + t_2 - t_3 - t_4| \leq 1 : t_i \in W\}$.

Theorem 2 (Heath-Brown [3]). *Suppose $(a_n)_{n\leq N}$ are complex valued such that $|a_n|\leq 1$, and that $S=\{t_1,\dots,t_R\}\subseteq[0,T]$ be a set of 1-separated points. We have*

$$\sum_{r,s\leq R}\Big|\sum_{n\sim N}a_n n^{-i(t_r-t_s)}\Big|^2\leq T^{o(1)}\big(R^2N+RN^2+R^{5/4}T^{1/2}N\big). \tag{2}$$

Furthermore, if $N\in[T^{2/3},T]$, we can ignore the last term.

Remark 2.1. *To prove a bound of this shape we can assume that the set S is T^{ε}-spaced. This is because we can split our 1-spaced set into $\approx T^{\varepsilon}$-many T^{ε}-spaced sets. Applying Cauchy–Schwarz, this only contributes a factor of T^{ε} to the bound, which is engulfed into the $T^{o(1)}$ term.*

First of all, we should put into perspective how strong this bound is. We have seen the following (very strong) conjecture:

Conjecture 1 (Montgomery's large value conjecture). *Let $S\subseteq[0,T]$ be a 1-spcaed set. We have*

$$\sum_{r\leq R}\Big|\sum_{n\sim N}a_n n^{-it_r}\Big|^2\leq T^{o(1)}(N+R)N.$$

Applying this bound to the left-hand side of (2) gives

$$\sum_{r,s\leq R}\Big|\sum_{n\sim N}a_n n^{-i(t_r-t_s)}\Big|^2\leq T^{o(1)}(N+R)N\sum_{s\leq R}1\ll T^{o(1)}(R^2N+RN^2),$$

so Theorem 2 is essentailly optimal when $N\in[T^{2/3},T]$. The goal here is to provide a rough proof of Theorem 2 in this case. First, we outline three key ideas that work together to give the proof.

2.1 Key Idea 1

With averages of this form, we have

$$\begin{aligned}\sum_{r,s\leq R}\Big|\sum_{n\leq N}a_n n^{-i(t_r-t_s)}\Big|^2&=\sum_{n,m\leq N}a_n\overline{a_m}\sum_{r,s\leq R}(n/m)^{-i(t_r-t_s)}\\&=\sum_{n,m\leq N}a_n\overline{a_m}\Big|\sum_{r\leq R}(n/m)^{-it_r}\Big|^2.\end{aligned} \tag{3}$$

This relation allows one to prove the following lemma.

Lemma 2.1. *Suppose $(a_n)_{n\leq N}$ are complex valued and $(b_n)_{n\leq N}$ are real valued such that $|a_n|\leq b_n$. Then*

$$\sum_{r,s\leq R}\Big|\sum_{n\leq N}a_n n^{-i(t_r-t_s)}\Big|^2\leq\sum_{r,s\leq R}\Big|\sum_{n\leq N}b_n n^{-i(t_r-t_s)}\Big|^2$$

Adam Harper pointed out in the study group that this result is an example of a *Majorant principle*, and results of this kind are often very useful.

Proof. The Lemma follows by applying the triangle inequality to (3) and working backwards. □

This result is particularly important because it allows us to consider $a_n\equiv 1$ or, similarly (and technically advantageous) $a_n=n^{-1/2}$. Along these lines, we define

$$\mathcal{A}(N)\coloneqq\sum_{r,s\leq R}\Big|\sum_{n\sim N}n^{-1/2-i(t_r-t_s)}\Big|^2.$$

It will suffice for our purposes to prove that

$$\mathcal{A}(N)\leq T^{o(1)}(R^2+RN),$$

when $N\in[T^{2/3},T]$.

2.2 Key Idea 2

We are reduced to considering Dirichlet polynomials of the form

$$\sum_{n\sim N} \frac{1}{n^{1/2+it}},$$

which are (some splitting of) the partial sums of the Riemann zeta function. Poisson summation tells us that (for say, f a Schwartz function),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \widehat{f}(n),$$

where $\widehat{f}$ denotes the Fourier transform of f. Poisson summation tells us that

$$\sum_{n\sim N} \frac{1}{n^{1/2+it}} \approx \chi(1/2+it) \sum_{n\sim t/2\pi N} \frac{1}{n^{1/2-it}}.$$

Here $\chi(s)$ is the function appearing in the functional equation of the Riemann zeta function, that is, $\zeta(s) = \chi(s)\zeta(1-s)$. It is important to note that $|\chi(1/2+it)| = 1$, so that taking absolute value, the above implies

$$\Big|\sum_{n\sim N} \frac{1}{n^{1/2+it}}\Big| \approx \Big|\sum_{n\sim t/2\pi N} \frac{1}{n^{1/2+it}}\Big|.$$

In particular, if $|t| \leq T$, and $N \geq T^{2/3}$, say, then the right hand side is a Dirichlet polynomial of length $\leq T^{1/3}$, which is much shorter than the Dirichlet polynomial on the right hand side. This should be very advantageous, since one should imagine that a shorter average would be genuinly smaller. This point cannot be emphasized enough: it is the key idea that makes the proof work.

There are a couple of technical details to mention. First of all, the above equation is clearly not true when t is very small. This is why we consider a T^ε-spaced set, as mentioned in Remark 2.1. Once the diagonal is removed, each term $\sum_{n\sim N} a_n n^{-i(t_r-t_s)}$ will have a sufficiently large twist. Secondly, from a technical perspective, it is quite annoying to have the polynomial length depending on the size of t (or $t_r - t_s$, in our case). If one uses a more compelx-analytic approach, one can actually prove the following lemma, which overcomes this annoyance.

Lemma 2.2. *Let $h = \log T$, and let $b_n = e^{-(n/2N)^h} - e^{-(n/N)^h}$. Then for $T^\varepsilon < |t| \leq T$, we have*

$$\Big|\sum_n b_n n^{-it}\Big| \ll N^{1/2} \int_{-h^2}^{h^2} \Big|\sum_{n\leq T/\pi N} n^{-1/2+i(t+\tau)}\Big| d\tau$$

One should just think of the coefficients b_n as a smoothing of $\mathbf{1}_{(N,2N]}(n)$, whose Mellin-transforms are (dilations of) the Gamma function, which has very good decay away from the real line. The proof goes by applying Mellin inversion to the coefficients b_n, shifting the contour left, then using the functional equation to flip the zeta-function. One then writes the zeta function in terms of its Dirichlet series and splits it at $n = T/\pi N$. See [5][Lemma 3] for more details (note that with $|t| > T^\varepsilon$, the decay from the gamma function is good enough that the constant term there disappears.).

2.3 Key Idea 3

The final idea will will make use of is the fact that longer Dirichlet Polynomial averages should be larger.

Lemma 2.3 (White lie lemma)**.** *$\mathcal{A}(N)$ is an increasing function of N.*

While this lemma is not exactly true, it is morally correct. For a precise statement, see [3, Lemma 2]. Importantly, the lemma delivers the following corollary, which, in conjunction with the previous idea, will allow us to relate $\mathcal{A}(N)$ to $\mathcal{A}(N)^{1/2}$.

Corollary 2.1. *If $N \in (T^{2/3}, T]$, we have*

$$\mathcal{A}(T/\pi N) \leq T^{o(1)} R \mathcal{A}(N)^{1/2}.$$

Proof of corollary. Applying Cauchy–Schwarz first, we have

$$\begin{aligned}
\mathcal{A}(T/\pi N) &= \sum_{r,s\leq R} \Big| \sum_{n\sim T/\pi N} n^{-1/2-i(t_r-t_s)} \Big|^2 \\
&\leq R \Big(\sum_{r,s\leq R} \Big| \sum_{n\sim T/\pi N} n^{-1/2-i(t_r-t_s)} \Big|^4 \Big) \\
&\leq T^{o(1)} R \Big(\sum_{r,s\leq R} \Big| \sum_{n\sim (T/\pi N)^2} n^{-1/2-i(t_r-t_s)} \Big|^2 \Big)^{1/2} \\
&\leq T^{o(1)} R \mathcal{A}(T^2/N^2),
\end{aligned}$$

where the third line follows by squaring out the fourth moment and appling Key Idea 1 (more specifically, we apply Lemma 2.1, making use of the divisor bound $\max_{n\sim (T/\pi N)^2} d(n) = T^{o(1)}$). Seeing as $N \in (T^{2/3}, T]$, it follows from Lemma 2.3 that we have $\mathcal{A}(T^2/N^2) \leq \mathcal{A}(N)$. This gives the corollary. □

2.4 Proof of Theorem 2

Rough Proof. As promised, we now make use of these three ideas to show how one proves Theorem 2 in the limited case where $N \in (T^{2/3}, T]$. We begin by replacing our coefficients (say $a_n = \mathbf{1}_{n\sim N} n^{-1/2}$) by the smoother coefficients $b_n = N^{-1/2}(e^{-(n/2N)^h} - e^{-(n/N)^h})$, where $h = \log T$. Since $a_n \ll b_n$, we can make use of Lemma 2.1. This gives

$$\mathcal{A}(N) \leq \frac{1}{N} \sum_{r,s\leq R} \Big| \sum_n b_n n^{-i(t_r-t_s)} \Big|^2.$$

We now remove terms where $r = s$ to find that

$$\mathcal{A}(N) \ll \frac{R}{N} \Big| \sum_n b_n \Big|^2 + \frac{1}{N} \sum_{\substack{r,s\leq R \\ r\neq s}} \Big| \sum_n b_n n^{-i(t_r-t_s)} \Big|^2.$$

One can verify that $\sum_n b_n \leq N$, which should not be surprising seeing as they were chosen to be a smooth approximation to $\mathbf{1}_{N,2N}(n)$. Additionally, we apply Lemma 2.2 to the second term to deduce that

$$\mathcal{A}(N) \ll RN + \sum_{\substack{r,s\leq R \\ r\neq s}} \Big(\int_{-h^2}^{h^2} \Big| \sum_{n\leq T/\pi N} n^{-1/2+i(t_r-t_s)+i\tau} \Big| d\tau \Big)^2.$$

Having applied our lemma, we can add back in the terms where $r = s$ We can also apply Cauchy–Schwarz to the integral to deduce that

$$\mathcal{A}(N) \ll RN + h^2 \int_{-h^2}^{h^2} \sum_{r,s\leq R} \Big| \sum_{n\leq T/\pi N} n^{-1/2+i(t_r-t_s)+i\tau} \Big|^2 d\tau.$$

By Lemma 2.1, we can upper bound the above integrand by the case when $\tau = 0$. This allows one to lose the (undesirable) dependence on τ. One can then break the sum over n into dyadic integvals, apply Cauchy–Schwarz, and apply Lemma 2.3 to deduce that

$$\mathcal{A}(N) \ll RN + T^{o(1)}\mathcal{A}(T/\pi N).$$

Finally, one can apply Corollary 2.1 to obtain a copy of $\mathcal{A}(N)$ on the right-hand side. This gives

$$\mathcal{A}(N) \ll RN + T^{o(1)}R\mathcal{A}(N)^{1/2}$$

It follows (for example, bu completing the square and applying the reverse triange inequality) that

$$\mathcal{A}(N) \ll T^{o(1)}(RN + R^2),$$

as intended. □

3 Application to large value sets with high additive energy

We now give a very brief sketch of how we use Theorem 2 to handle large value sets with high additive energy. See Guth and Maynard [1, Section 11] for more detail. Recall that, for W a set of real numbers, we define the additive energy as

$$E(A) = \#\{|t_1 + t_2 - t_3 - t_4| \leq 1 : t_i \in A\}.$$

We begin with $S \subseteq [0, T]$ a 1-spaced set. Suppose that $(b_n)_{n\leq N}$ are complex numbers with $|b_n| \leq 1$, and $S_V \subseteq S$ are the points $t \in S$ such that

$$D(t) := \Big| \sum_{n\sim N} b_n n^{it} \Big| \geq V.$$

The main idea is that Dirichlet polynomials don't fluctuate too much over intervals of length ≈ 1 (see [1, Lemma 11.3], for example), so that if $|t_1 + t_2 - t_3 - t_4| \leq 1$ then we have $D(t_4) \approx D(t_1 + t_2 - t_3)$. More precisely, the argument goes as follows:

$$\begin{aligned}
E(S_V) &= \sum_{t_1,t_2,t_3,t_4\in S_V} 1 \leq V^{-2} \sum_{t_1,t_2,t_3\in S_V} |D(t_4)|^2 \\
&\leq V^{-2} \sum_{t_1,t_2,t_3\in S_V} |D(t_1 + t_2 - t_3)|^2 \\
&= V^{-2} \sum_{n,m\sim N} a_n \overline{a_m} \Big(\sum_{t\in S_V} (n/m)^{it} \Big)^2 \Big(\sum_{t\in S_V} (m/n)^{it} \Big) \\
&\leq V^{-2} \sum_{n,m\sim N} \Big| \sum_{t\in S_V} (n/m)^{it} \Big|^3.
\end{aligned}$$

Applying the trivial bound for one of the sums over t, we have

$$\begin{aligned}
E(S_V) &\ll V^{-2}|S_V| \sum_{n,m\sim N} \Big| \sum_{t\in S_V} (n/m)^{it} \Big|^2 \\
&= V^{-2}|S_V| \sum_{t_1,t_2\in S_V} \Big| \sum_{n\sim N} n^{-i(t_1-t_2)} \Big|^2.
\end{aligned}$$

This is precisely the mean square appearing in Theorem 2. Applying this theorem, we have

$$E(S_V) \leq T^{o(1)}(|S_V|^3 NV^{-2} + |S_V|^2 N^2 V^{-2}),$$

whenever $N \in (T^{2/3}, T]$. Now, if the additive energy of S_V is near-maximal, say $E(S_V) \geq T^{o(1)}|S_V|^3$, and if $V > N^{1/2+\varepsilon}$ (this is the range of V that is of most interest), then the first term on the right-hand side is redundant, and we have

$$|S_V|^3 \leq T^{o(1)}|S_V|^2 N^2 V^{-2}.$$

In other words, $|S_V| \leq T^{o(1)} N^2 V^{-2}$ whenever $V > N^{1/2+\varepsilon}$. This estimate is the best that one could hope for: Montgomery's conjecture gives the same bound.

References

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